

The Particle Model — Paper B

MEDIUM • NON-CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says explain, your answer must talk about particles — their arrangement, their movement, the spaces between them, or the forces between them.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – HEATING AND COOLING

1. A solid is heated steadily and its temperature recorded every two minutes.

Time (min):	0	2	4	6	8	10	12
Temp (°C):	30	54	80	80	80	96	118

(a) State the melting point of the substance, and how you can tell. [2]

(b) State how long the substance took to melt. [1]

(c) Describe what is happening to the particles between 4 and 8 minutes. [2]

2. A different substance boils at 78 °C and melts at –114 °C. State what state it is in at room temperature (20 °C), and explain how you know. [3]

- 3.** Explain why the temperature of a pure substance does not rise while it is melting, even though the heater is still switched on. [3]

SECTION 2 – MASS AND MOVEMENT

- 4.** A sealed flask containing water is weighed. It is then warmed until all the water has turned to steam, and weighed again.

(a) State what happens to the reading on the balance. [1]

(b) Explain your answer in terms of particles. [2]

(c) State what would be different if the flask had been left open. [1]

- 5.** A drop of ink is placed at the bottom of a tall glass of still water. Two days later the whole glass is evenly coloured, although nobody stirred it.

(a) Name the process. [1]

(b) Explain how the ink reached the top. [2]

(c) State one change that would make it happen faster. [1]

- 6.** Smoke is viewed in a small glass cell under a microscope, and the specks are seen to jiggle about in random directions.

(a) Name what is being observed. [1]

(b) Explain what is causing the specks to move. [2]

SECTION 3 – SPEED AND PRESSURE

7. Explain why a smell spreads across a warm room faster than across a cold one. [3]

8. Name the change of state in which a solid turns directly into a gas, and give one example. [2]

9. A sealed balloon is put into a fridge and is found to shrink slightly. Explain this in terms of particles. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.